QP CODE:25900399

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCFTQ201 - FOOD ANALYTICAL INSTRUMENTATION
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap) Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short Answer Questions Answer any 5 questions Each question carries 2 marks

1. What are the applications of chromatography in food industry? [U] / [CO1]
Chromatography plays a crucial role in the food industry for quality control, safety, and product development**.**

2. Define the mobile phase in GC and give examples. [U] / [CO2]
In gas chromatography (GC), the mobile phase is an inert gas, called the carrier gas that transports the sample through the column. Common examples include helium, nitrogen, and hydrogen. The carrier gas moves the analytes through the stationary phase, where they are separated based on their different physical and chemical properties

3. What are the benefits and limitations of starch gel electrophoresis? [K] / [CO5]
Starch gel electrophoresis is a technique used for separating biomolecules, particularly proteins and nucleic acids, based on their size and charge. Benefits: High Resolution: Provides excellent separation of closely related proteins or nucleic acids.
Limitations: Limited Size Range, Longer Run Times, Difficult Staining

4.Define electrophoresis. Who pioneered the technique of electrophoresis.[K][CO5]
Electrophoresis is a laboratory technique that uses an electric field to move charged molecules in a fluid or gel, separating them based on their size, charge, or binding affinity. It is used to separate and analyze molecules like DNA, RNA, and proteins. The Swedish biochemist Arne Tiselius pioneered the technique of electrophoresis in the 1930s

5. What is the role of centrifugation in dairy industry? [U] / [CO5]
Centrifugation plays a crucial role in the dairy industry by separating components based on density, enhancing product quality and efficiency.
Separation of Cream and Milk, Clarification of Milk, Standardization of Milk

6. What is nuclear magnetic resonance? [U] / [CO4]

Nuclear magnetic resonance (NMR) is a physical phenomenon in which nuclei in a strong constant magnetic field are disturbed by a weak oscillating magnetic field and respond by producing an electromagnetic signal with a frequency characteristic of the magnetic field at the nucleus

7. What is the principle of ICP-AES? ? [U] / [CO4]

ICP-AES, or Inductively Coupled Plasma Atomic Emission Spectroscopy, is an analytical technique used to detect and quantify elements in a sample.

Principle of Operation: Sample Introduction, Plasma Generation, Excitation of Atoms and Detection (5x2=10 marks)

Part B

Short Essay Questions Answer any 5 questions Each question carries 4 marks

8. Add a note on microchip electrophoresis. [K] / [CO5]

Microchip electrophoresis is a technique used to separate proteins, nucleic acids, and charged molecules in an electric field, which enhances speed and simplicity compared to traditional methods**.**

9. What is the difference between horizontal and vertical electrophoresis? [U] / [CO5]

Horizontal and vertical electrophoresis are techniques used to separate biomolecules, primarily nucleic acids and proteins, based on size and charge.

Orientation: Horizontal Electrophoresis and Vertical Electrophoresis

10. List out the factors affecting fluorescence. [K] / [CO4]

Factors affecting fluorescence include the intrinsic molecular structure (such as conjugation and rigidity) and the external environment (like temperature, viscosity, pH, and the presence of quenchers and oxygen). Other factors include concentration, the excitation wavelength, and the intensity of the incident light

11. What is a Nernst Glower? [U] / [CO4]

A Nernst glower is a ceramic rod made of rare-earth oxides (like zirconium, yttrium, and erbium oxides) that emits infrared radiation when heated to high temperatures by an electric current.

12. In a TLC experiment, two components A and B moved 7 cm and 9 cm respectively from the origin. The solvent front moved 10 cm. Calculate the Rf values for components A and B. [A] / [CO2]

Rf=Distance travelled by solute/Distance travelled by solvent front

13. Describe the phenomenon of interaction radiation with matter? [U] / [CO3]

The interaction of radiation with matter is the process where radiation transfers energy to or changes direction in a material. This can result in absorption, where the radiation's energy is completely deposited in the matter; scattering, where the radiation changes direction, often losing some energy; or transmission, where the radiation passes through the material without interacting. **Electromagnetic radiation (e.g., X-rays, gamma rays)**

14. Give an overview of different detectors employed in HPLC? [U] / [CO2]

HPLC detectors are categorized as either selective (for specific solute properties) or universal (for bulk properties), with common examples including UV/Vis, fluorescence, mass spectrometry, refractive index, and electrochemical detectors

(**5x4=20 marks)**

Part C

Long Essay Questions. Answer any 2 questions. Each question carries 10 marks

15. Explain the principle, working, and applications of chromatography paper [U] / [CO2]

Paper chromatography separates a mixture based on its components' different solubilities and affinities for the stationary and mobile phases. The process involves a stationary phase of water trapped in the paper and a mobile phase of a solvent that moves up the paper by capillary action. Components of a mixture spotted on the paper are carried up by the solvent at different rates, leading to their separation into distinct spots at different heights. Applications include separating inks, analyzing dyes in food, testing for drug doping, and identifying pigments in plants

16. Explain techniques and working of atomic absorption spectroscopy. [U] / [CO4]

Atomic absorption spectroscopy (AAS) is an analytical technique used to measure the concentration of a specific metal in a sample by atomizing it and then passing a beam of light through the resulting atomic vapor.

17. Write briefly about differential, rate zonal, and isopycnic centrifugation methods. [U] / [CO5]

Differential centrifugation: This technique separates particles primarily by size and density using a series of increasing centrifugal forces. It is a simple and common method for initial purification and isolating subcellular components.

Rate-zonal centrifugation separates particles by their sedimentation rate (size and shape), while isopycnic centrifugation separates them by their buoyant density, independent of size. Rate-zonal uses a shallow gradient and shorter run times, best for separating particles that differ in size but not density, like organelles. Isopycnic uses a steep gradient and longer run times, ideal for separating particles based on density.

18. Describe the different modes of centrifugation. [U] / [CO5]

The main modes of centrifugation are differential centrifugation and density gradient centrifugation, with the latter often broken down into rate zonal and isopycnic techniques. Differential centrifugation separates components by spinning samples at increasing speeds, while density gradient centrifugation separates based on density using a prepared gradient medium, which can either be separated by speed (rate zonal) or by the density at which components reach equilibrium (isopycnic)

[2x10 = 20]